A45U RMU-2026

Rack-Mountable AC Load Testing — Engineered for Data Centers

4 kW Rack Mount Load Bank

Direct server substitution for heat & power load simulation

WHY OUR RACK LOAD BANKS?

Server placement and HVAC/PDU upgrades need real load.

Guesswork on white space = hot spots, under-utilized capacity, or failed commissioning.

The load bank delivers 4 kW (or 2 kW) of precision AC resistive load in a 19" rack-mount package — drop-in replacement for a server, switch, or modem bank.

Simulate front-to-back, hot-aisle/cold-aisle, or base-to-top airflow without risking expensive IT equipment.

Modular from 1–24 kW. Stack or distribute units to match real rack densities and verify power and cooling upgrades.

INSTANT BENEFITS

- 19" rack mountable — Direct substitution of computer, data switch, modem bank, or server.
- 4 kW @ 208 VAC / 2 kW @ 120 VAC — Two isolated channels; (2) 2 kW steps or (2) 1 kW steps.
- Front-to-back airflow — Cool air drawn from cold aisle, hot air discharged out the back.
- Modular 1–24 kW — Assemble multiple units for realistic rack load profiles.
- Any plug style — Rear terminal blocks + selector switch; optional cord assemblies.
- Safe commissioning — Validate HVAC, PDU, and UPS upgrades without live server risk.
- Compact 26" × 19" × 9" — Fits standard racks; 44 lbs.
- NET Airflow 474.6 CFM — 237.3 CFM each channel.
- Approved for 240 V, 4 kW (3-phase with multiple units).
- Remote Load On — Rear-panel connection for remote control.



19" Rack-Mount 4 kW AC Load Bank

GREAT FOR TESTING

- Individually cooled rack designs
- HVAC capacity and airflow verification
- PDU and UPS load / upgrade validation
- Server placement and power-density planning
- Hot-aisle / cold-aisle thermal studies
- Data center commissioning and white-space optimization
- Front-to-back or base-to-top airflow patterns
- Rack-level heat load simulation (1–24 kW)
- Cooling redundancy and fail-over testing
- Power distribution and branch-circuit loading
- CRAC/CRAH unit capacity and response testing
- Containment system effectiveness validation
- Transient load and step-change response studies
- Thermal imaging and hot-spot identification

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LOAD CONFIGURATION

Total Load	Voltage	Step Resolution	Notes
4 kW	208 VAC	2 kW	(2) isolated channels
2 kW	120 VAC	1 kW	(2) × 1 kW with common return

ELEMENT RATINGS (per channel)

Configuration	Voltage	Resistance	Current / Power
120 VAC mode	120 VAC	14.4 Ω	≈8.33 A / 1000 W
208 VAC mode	208 VAC	21.6 Ω	≈9.62 A / 2000 W

Base elements are two 15 Ω resistors with multiple tap locations. Internal 24 VDC supplies for fans and selector relays are powered from the load inputs; ohmic values are adjusted so the additional draw is transparent.

CONTROLS & PROTECTION

- Rear terminal block inputs
- Rear selector switch (208 VAC / 120 VAC)
- Front-panel double-pole circuit breaker
- Front-panel Load On (with remote rear connection)
- Internal 24 VDC for fans & relays

FEATURES

- 19” rack-mount form factor
- Front-to-back airflow (cold aisle in / hot aisle out)
- Optional plug & cord assemblies
- Customer supplies racks, blanking panels, shelves, brackets
- ≈80 dB sound level

DIMENSIONS

- Length: 26” (66 cm)
- Width: 19” (48.3 cm)
- Height: 5U 9” (22.7 cm)
- Weight: 44 lbs (20 kg)

SHIPPING DIMENSIONS

- Length: 33” (83.8 cm)
- Width: 23” (58.4 cm)
- Height: 15” (38.1 cm)
- Weight: 58 lbs (26.3 kg)

SPECIFICATIONS

- Unit airflow: 474.6 CFM
- Fans: ETRI Model 148DH, 24 VDC, 237.3 CFM each
- Sound level: approx. 80 dB
- Channels: 2 isolated
- Protection: double-pole circuit breaker
- Remote Load On capability

APPLICATION NOTES

- Ideal for individually cooled rack testing.
- Assemble 2, 4, 6, 8, or 12 kW (or up to 24 kW) per rack.
- Verify kW-per-rack loads to maximize white space.
- Suitable for front-to-back, hot/cold aisle, or base-to-top patterns.
- Customer provides mechanical hardware and blanking panels.

Engineered for Performance. Delivered with Precision.

Load Bank Innovation · Commissioning Excellence · Power Reliability

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